BevoTag Project Definition: Rubric Review

Oct 15, 2026 · Li Shi · Live version with status dropdowns: claude.ai/code/artifact/dfe24570-ae46-467c-b283-620e5008a698

Verdict and how to use this doc

The revised Project Definition would most likely earn **Complete: Meets Basic Expectations** today. The formatting and structure problems from the Writing TA feedback are mostly fixed. What keeps it from **Exceeds** is content: missing specifications, a few citations that may not support their claims, and places where the doc commits to a specific design.

This review covers `docs/project-definition/main.tex` as pushed to `main` (commit 9c0a6c9). It checks the draft against `Project Definition rubric.pdf`, the assignment description, and the ECE Style Guide. The ratings below are an estimate, not a TA's grade.

Issues fall into three groups:

- **Needs team discussion:** decisions or numbers the group has to agree on, ideally at the next meeting. Each comes with a question and a suggested answer.
- **Needs attention:** real fixes that one person can make without a group decision. Assign an owner.
- **Trivial fixes:** a few minutes each, for anyone.

Use the Status column to track progress, and comment on any row you disagree with.

Meeting agenda

A 40-minute meeting covers the 11 decisions. The other 19 fixes only need an owner and a deadline, not discussion.

1. **Verdict and scorecard** (3 min): agree on the target of reaching Exceeds.
2. **Specification numbers**, D1, D2, D3, D11 (10 min).
3. **Scope and framing**, D4, D5, D6 (8 min).
4. **Standards and ethics**, D7, D8 (8 min).
5. **Team roles and mentor input**, D9, D10 (5 min).
6. **Owners and deadline** (5 min): confirm owners for A1–A10 and T1–T9, and set the internal deadline. Team policy puts it 48 h before the course deadline.

Write each outcome in the Decision column as you go, and set Status to Decided.

Before the meeting:

- ☐ Everyone: read the Needs team discussion section and comment on any row you disagree with.
- ☐ Ethan: bring notes from mentor meetings (D10) and the final due date (T1).
- ☐ Esteban: bring rough ESP32 current-draw numbers for the battery-life estimate (D1).
- ☐ Li: check whether UT floor plans may sit in a public repo (D8).

Rubric scorecard

Content and team organization are the weakest areas. Structure, visuals, and formatting are close to the Exceeds level.

Rubric area	Estimate	Main reason
Structure: Introduction	Above Average	Purpose and a summary of ideas are there. It never says where the project stands now, even though a data logger and survey collector already exist.
Structure: Conclusion	Above Average	Restates the purpose, assesses risk realistically, and names next steps. The next steps are out of date (see A10).
Structure: Sections	Above Average	Every section has an introductory paragraph. The Problem Statement and Design Problem Description intros both just preview topics.
Structure: Paragraphs	Above Average	Topic sentences come first. A few paragraphs are lists written as sentences.
Organization: Body	Average	GNSS trackers' weaknesses are covered twice. “How BevoTag Determines Its Location” reads as a chosen design.
Organization: Paragraphs	Above Average	Each paragraph supports a single idea.
Content: Problem Statement	Average	Specs are missing battery life, update rate, floor accuracy, size, and the software environment. Some citations may not support their claims. No mentor input is recorded.
Content: Team Organization	Average	Only one team-wide role. Four members' expertise and coursework read the same.
Visual Aids and Appendixes	Above Average	The I/O diagram and responsibilities table are there and referenced in the text. Figure 2's caption carries a footnote, and the appendix relies on a weak standard (IEEE 1451).
Mechanics, Grammar, Style	Above Average	Some point-of-view and line-break slips, plus pronouns that need confirming (see T2).
Formatting	Above Average	Matches the Style Guide and memo template. Red placeholder text must be removed before submission.

Needs team discussion

Eleven items need a group decision before anyone edits. Most are specification numbers that only the team, or Dr. Vikalo, can set. They are grouped in agenda order. The first group matters most for the rubric's "thorough set of specifications" criterion.

1. Specification numbers (10 min)

#	Issue	Question for the group	Suggested answer	Suggested owner	Decision	Status
D1	No power or battery-life spec, although power is the first design challenge and the conclusion stresses low power	What battery life and reporting interval can we promise?	Add two rows to Table 4: "Battery life ≥ __ days at one report every __ min" and "Location update interval while moving ≤ __ s." Estimate from the ESP32's measured current draw, then confirm with Dr. Vikalo.	Esteban		Open
D2	No size, mass, or cost spec; the deliverables only say "compact"	How small, light, and cheap must the tag be?	Add rows for maximum dimensions (mm), mass (g), and unit parts cost ($).	Esteban		Open
D3	Test buildings are multi-story, but no spec covers floor identification	Does "room identification accuracy" include getting the floor right?	Add "Floor identification accuracy > __%" to Table 4, and state that room accuracy is measured across floors.	Andrew		Open
D11	The revision proposed values for the draft's "???" cells (red in the PDF)	Accept 0–40 °C, 0–2 m/s walking speed, and an IMU rate ≥ 100 Hz?	Confirm or change the values. Move the IMU rate out of the performance table (see A6).	Li		Open

2. Scope and framing (8 min)

#	Issue	Question for the group	Suggested answer	Suggested owner	Decision	Status
D4	No software, computational, or network environment; the assignment requires one for software-heavy projects	Where does the backend run, and which platforms must the app support?	Add Table 3 rows for the server host (UT VM, cloud, or a laptop for the demo), supported browsers or phone OS, and any device registration utexas-iot requires (ref [4]).	Abhirit		Open
D5	The doc presents the EKF, ESP32-S3, and server-side estimation as decided; the guideline says "do not provide a specific proposed design," and Design Challenges calls the tag-versus-server split open	Present the approach as a decided design, or as a technical premise?	Rename "How BevoTag Determines Its Location" to "Technical Premise." Say estimation runs "on the tag or the server," relabel Figure 1's server box "Location processing," and keep ESP32-S3 only as an example.	Ethan		Open
D6	Deliverables, Table 2, and Figure 2 promise a "3D map," but the survey work uses 2D floor plans per floor	Will we deliver a true 3D map?	If not, change "3D map" to "multi-floor campus map" in all three places.	Andrew		Open

3. Standards and ethics (8 min)

#	Issue	Question for the group	Suggested answer	Suggested owner	Decision	Status
D7	IEEE 1451 is a weak fit (few IMUs or ESP32 projects follow it), and the cited "1451.2-2014" edition looks wrong	Keep IEEE 1451 or replace it?	Replace it with IEC 62133-2 (lithium battery safety), which also backs the battery-safety ethics paragraph. Consider the FCC RF exposure rules for portable devices (47 CFR §2.1093) or the USB Type-C specification. Keep the appendix at about four standards.	Ethan		Open

#	Issue	Question for the group	Suggested answer	Suggested owner	Decision	Status
D8	Ethics only promises to "evaluate safeguards" against stalking, and it skips intellectual property (one of the guideline's four example topics)	What concrete anti-misuse measure do we commit to? How do we handle the UT floor plans in our public GitHub repo?	Commit to owner-only access plus one concrete safeguard, or declare it out of scope with a reason. Add an IP paragraph on floor-plan use rights and open-source licenses. Make the repo private, or remove the plan PDFs until UT Facilities clears them.	Li		Open

4. Team roles and mentor input (5 min)

#	Issue	Question for the group	Suggested answer	Suggested owner	Decision	Status
D9	Only one team-wide role (Project Lead); the guideline suggests others. The original draft also contradicted itself on Andrew's and Abhirit's roles	Who takes documentation lead, test lead, and repo maintainer? Is Andrew on localization and backend, with Abhirit on the app?	Name two or three roles beyond Project Lead. The documentation lead owns Style Guide checks before each submission. Confirm the Andrew/Abhirit split the revision assumed.	Ethan		Open
D10	No mentor input recorded; the guideline says this section records what you learned from your mentor	What has Dr. Vikalo asked for (accuracy, demo buildings, deliverables)?	Add two or three sentences to Background or Deliverables stating the mentor's expectations, cited as an IEEE "private communication."	Ethan		Open

Needs attention

Ten fixes need real work but no group decision. At the meeting, only confirm each owner. A1–A3 carry the most risk: a citation that doesn't support its claim can cost credibility with both TAs.

#	Issue	Suggested solution	Suggested owner	Status
A1	Three citations may not support the claims they back. [3] Jiang & Yuan is a device-free positioning paper, used for "RSSI-only error of 5–15 m." [5] Mukherjee covers radar-inertial odometry on ground robots, used for "IMU error builds within seconds." [2] Yoder & Humphreys is used for "low-cost motion sensors drift."	Open each paper and find the exact passage. Where it doesn't match, replace the source or soften the claim. Add page numbers when citing a specific fact (Style Guide 3.1, rule 5). Keep [2] for multipath, where it fits.	Ethan	Open
A2	Uncited claims: tracking tags used for stalking, the Samsung SmartTag, and "students routinely lose valuable items"	Cite a news or industry source for misuse (Apple's and Google's unwanted-tracking alerts are documented). Cite Samsung's product page. Find UTPD lost-and-found or campus theft numbers to make the need concrete.	Abhirit	Open
A3	No CSI-localization research is cited; Existing Solutions covers only products	Cite two or three papers, such as SpotFi (Kotaru et al., SIGCOMM 2015) and published ESP32 CSI tools, and reuse them in the Prior Art Report.	Andrew	Open
A4	Table 1 doesn't set values: "Device-dependent" and "N/A" fill most cells, and "Telemetry" is internal to the system	Give an RSSI range (about −100 to −30 dBm), the CSI sub-carrier count the chip reports (check the ESP-IDF CSI docs), and IMU ranges (±g, ±°/s) from the chosen part. Drop the Telemetry row and mark GNSS optional.	Li	Open
A5	Two Table 2 criteria can't be tested: "Display matches latest estimate" and "Standard map controls"	Rewrite them as measurable criteria, e.g., "Map shows a new estimate within 5 s" (ties to Table 4), "Pan and zoom work in Chrome and Safari," and "Tracks at least __ tags."	Abhirit	Open
A6	Table 4 includes "IMU sampling rate," a design parameter rather than a performance target, and lacks availability	Move the IMU rate to Table 1. Add "Availability: percent of test time with a valid position ≥ __%."	Andrew	Open
A7	Repetition: the Problem Statement intro only previews topics, and GNSS trackers' indoor loss and power drain appear twice	Make the Problem Statement intro state the problem in two or three sentences. Cut the "Dedicated GNSS trackers" sentence from Limitations, since the GNSS subsection already covers it.	Documentation lead (D9)	Open

#	Issue	Suggested solution	Suggested owner	Status
A8	Table 5 doesn't tell members apart: Esteban and Li have identical expertise, and four members list the same three courses	Each member rewrites their own row and paragraph with specific projects, internships, and the courses that set them apart.	Each member	Open
A9	The Introduction never states the current stage of development, which the memo template asks for	Add one sentence: firmware already logs Wi-Fi RSSI and CSI, and a browser survey collector maps EER, with ASE and ECJ floor plans imported.	Li	Open
A10	The Conclusion's next steps are out of date: it says the team will "complete a data logger" and survey EER, but both have already started	Replace them with the real next steps, such as collecting the EER survey, measuring a baseline accuracy, and selecting components.	Ethan	Open

Trivial fixes

Nine small edits, each a few minutes. T1 and T7 must be done before submission, or the red placeholder text will show in the PDF.

#	Issue	Fix	Suggested owner	Status
T1	The memo date and Writing TA name are red placeholders	Fill in the final due date and the Writing TA's name.	Ethan	Open
T2	Pronouns: Personnel uses "he" for Ethan (from the original draft) and for Abhirit (added in the revision without checking)	Confirm each member's pronouns, or rephrase those sentences without them.	Each member	Open
T3	"Our team's" in the Introduction's first sentence; the rest of the doc says "the team"	Change it to "the team's."	Li	Open
T4	"Wi-Fi" breaks across lines in places (e.g., page 8)	Add a non-breaking Wi-Fi macro in main.tex.	Li	Open
T5	Figure 2's caption ends with "(*optional)"; a caption should be a title only	Move the note into the figure as a small legend line: "* optional."	Li	Open
T6	Figure 2 is 1.8 pt wider than the text block	Shrink the diagram by about 1 mm.	Li	Open
T7	Red TBD text remains in the memo heading, Table 3, Table 4, and Appendix Table 8	Resolve D7, D11, and T1, then search main.tex for `\tbd` until none remain.	Documentation lead (D9)	Open
T8	Tables use 11 pt text; the Style Guide specifies 12 pt Times without mentioning tables	Ask the Writing TA. If 12 pt is required, remove `\small` from the table environment.	Ethan	Open

#	Issue	Fix	Suggested owner	Status
T9	Standards references [6]–[10] were written from memory	Check each title, number, and year against the actual document. For example, IEEE 802.11-2020 was published in 2021.	Ethan	Open

Course resources still needed

The review used the rubric, assignment description, ECE Style Guide, memo template, IEEE citation reference, and the Discord summary of the Writing TA's feedback. Five more sources would sharpen it:

Resource	Where	Why it matters
Writing TA's inline comments on the preliminary submission	Canvas, on the submitted PDF	The rubric says those comments mark the specific places to fix. The Discord summary may miss some.
Technical TA feedback	Canvas or email	Likely covers specs and scope, which is where this draft is weakest (D1–D6).
Notes from meetings with Dr. Vikalo	Team Google Drive	Needed for D1 and D10: mentor expectations and spec targets.
EE 333T handouts on introductions and paragraph structure	Canvas or 333T notes	The TA said to “remember 333T stuff”; the rubric's paragraph criteria come from there.
An example Project Definition rated “Exceeds”	Canvas, if posted	Shows how deep and how long the TAs expect each section to be.